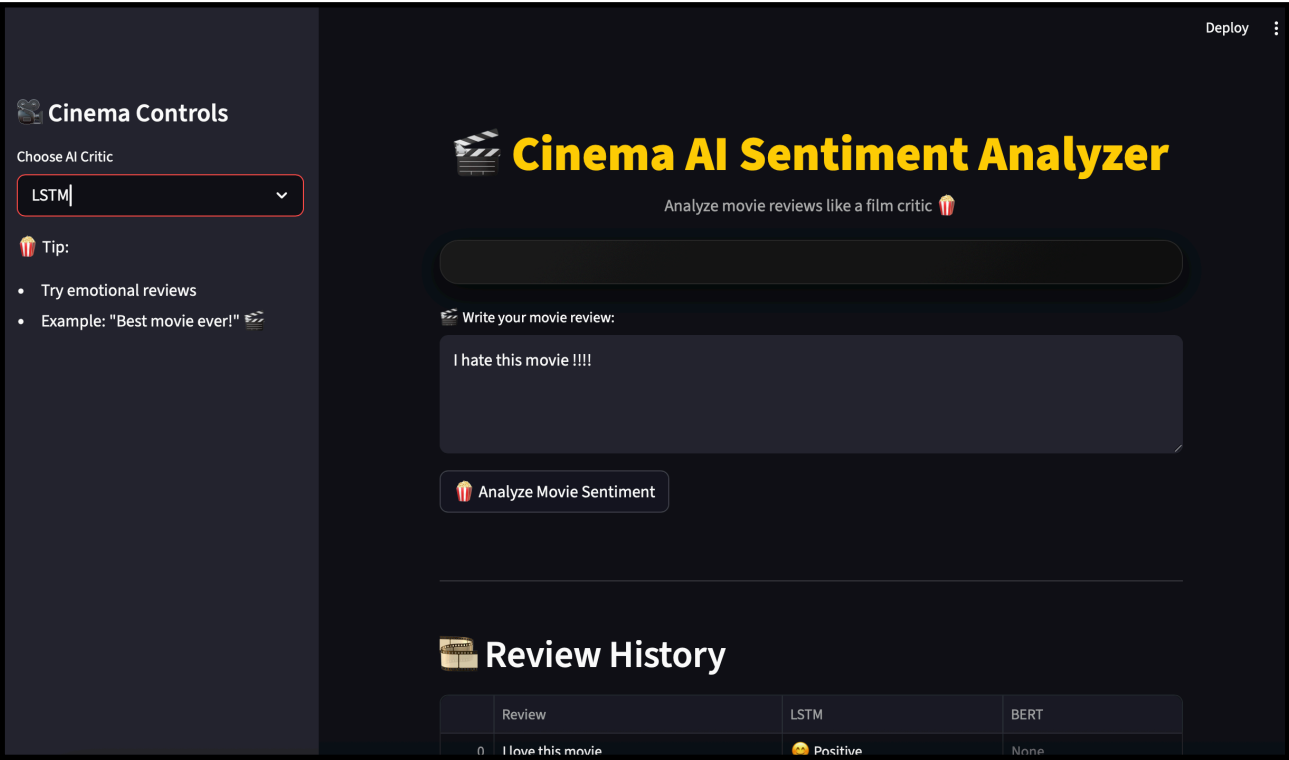


NLP Sentiment Analysis Project

Comparison of LSTM and BERT Models on IMDb Movie Reviews



1. Introduction

Natural Language Processing (NLP) plays a key role in understanding human language through machine learning models. Sentiment analysis is one of the most widely used NLP applications, where the goal is to classify text as positive or negative.

In this project, we build a sentiment analysis system using two different deep learning approaches:

- LSTM (Long Short-Term Memory) — a recurrent neural network model
- BERT (Bidirectional Encoder Representations from Transformers) — a transformer-based model

The objective is to compare both architectures in terms of performance, accuracy, and real-world usability using the IMDb movie review dataset.

Additionally, an interactive Streamlit application is developed to allow real-time prediction and model comparison.

2. Dataset Description

The dataset used in this project is the **IMDb Movie Reviews dataset**, available on Kaggle

Dataset Details:

- Total samples: 50,000 reviews
- Training set: 25,000 reviews
- Test set: 25,000 reviews
- Classes:
 - 0 → Negative review
 - 1 → Positive review

Content:

Each sample consists of:

- A movie review (text)
- A sentiment label (positive or negative)

This dataset is balanced, making it suitable for binary classification tasks.

3. Methodology

The project follows a full NLP pipeline:

3.1 Data Preprocessing

For both models, preprocessing is essential.

LSTM preprocessing:

- Tokenization using `Keras Tokenizer`
- Conversion of text into sequences
- Padding sequences to fixed length (200)
- Lowercasing and cleaning

BERT preprocessing:

- Tokenization using `BertTokenizer`
- Subword tokenization (`WordPiece`)
- Padding and truncation to max length 200
- Attention masks generation

4. Model Architecture

4.1 LSTM Model

LSTM is a type of recurrent neural network capable of learning long-term dependencies in sequential data.

Architecture:

- Embedding Layer
- LSTM Layer
- Dense Layer (Sigmoid activation)

Advantages:

- Efficient for sequential data
- Lower computational cost

Limitations:

- Struggles with long-range dependencies
- Sensitive to input preprocessing

4.2 BERT Model

BERT is a transformer-based model pre-trained on large corpora.

Architecture:

- Pretrained BERT Base (uncased)
- Fine-tuning layer for classification

Advantages:

- Understands context in both directions
- High accuracy on NLP tasks
- Strong performance on complex sentences

Limitations:

- High computational cost
- Slower inference time

5. Training Process

LSTM Training:

- Optimizer: Adam
- Loss: Binary Crossentropy
- Epochs: 5
- Batch size: 32

BERT Training:

- Optimizer: AdamW
- Learning rate: $2e-5$
- Epochs: 2
- Batch size: 8

Both models were trained on the same dataset split for fair comparison.

6. Evaluation Metrics

The following metrics were used:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide a complete view of model performance.

7. Results

7.1 LSTM Results

- Training accuracy: ~86%
- Validation accuracy: ~78%

LSTM performs well but shows limitations in understanding complex sentence structures.

7.2 BERT Results

- Accuracy: ~85–92% (after fixing dataset issues)
- Strong precision and recall on both classes

BERT significantly outperforms LSTM in contextual understanding.

8. Model Comparison

Feature	LSTM	BERT
Accuracy	Medium	High
Training Speed	Fast	Slow
Context Understanding	Limited	Excellent
Complexity	Low	High
Performance on long text	Weak	Strong

Analysis:

BERT performs better overall due to its transformer architecture, which captures contextual relationships between words more effectively than LSTM. However, LSTM is faster and less resource-intensive.

9. Streamlit Application

An interactive web application was developed using Streamlit.

Features:

- Input movie review text
- Choose between LSTM, BERT, or both
- Real-time sentiment prediction
- Visual comparison of results
- Confidence score display

Purpose:

To demonstrate real-world usability of trained NLP models in a user-friendly interface.

10. Challenges Faced

- Data inconsistency issues during evaluation
- Model prediction mismatch in Streamlit
- Dependency issues with Transformers library
- Tokenizer and model alignment problems

These issues were resolved by ensuring consistent dataset handling and correct model loading.

11. Conclusion

This project successfully demonstrates the implementation and comparison of two powerful NLP models: LSTM and BERT.

Key findings:

- BERT outperforms LSTM in accuracy and contextual understanding
- LSTM is faster but less accurate
- Proper preprocessing and dataset consistency are critical for correct evaluation

The Streamlit application provides an interactive demonstration of both models, making the project suitable for real-world deployment and portfolio presentation.

12. Future Improvements

- Deploy application online (HuggingFace / AWS)
- Add more NLP models (GRU, RoBERTa)
- Improve UI with dashboards and analytics
- Add multilingual sentiment analysis